

Algebra II with Trigonometry

Chapter 5.4

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401270_alg-2trig-h-chp-5-4-note-docx.pdf](#)

Problems

1. Evaluate exactly:

$$\sin\left(2 \arctan \frac{3}{4}\right) + \cos\left(2 \arcsin \frac{3}{5}\right).$$

2. Find the exact value of

$$\arctan \frac{1}{2} + \arctan \frac{1}{3} + \arctan \frac{1}{7}.$$

3. Let

$$\theta = \arcsin \frac{5}{13} + \arccos \frac{12}{13}.$$

Find $\tan \theta$ exactly.

4. A pole stands vertically on level ground. From point A , the angle of elevation to its top is 45° . After walking 20 meters directly away from the pole to point B , the angle of elevation becomes $\arctan \frac{2}{3}$. Find the height of the pole.

5. For $x \in [-1, 1]$, simplify exactly:

$$\sin(2 \arccos x) + \cos(2 \arcsin x).$$

6. Find all $\theta \in [0, \pi]$ satisfying

$$\arcsin(\sin \theta) = \arccos(\cos \theta).$$